Kailen Patel

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EDUCATION

University of Illinois at Urbana-Champaign **Expected May 2028**
Bachelor of Science in Computer Science *GPA: 3.33*

WORK EXPERIENCE

Infraly LLC **West Chicago, IL**
Server Technician *March 2024 - Present*

- Assemble servers with new motherboards, CPUs, RAM, storage, and cooling to expand computing capacity for client workloads
- Upgrade servers with higher-performance components or peripherals, including 10GbE NICs, which increase network capacity by up to 10x
- Provide 24/7 on-call support to quickly fix urgent server issues like component failures, maintaining 99.99% uptime

PROJECTS

Networking Homelab *July 2026 - Present*

- Configured WAN/LAN interfaces, DNS, DHCP, and firewall rules to support an expanding network of servers, client devices, and services
- Implemented WireGuard VPN, SSH, and MFA, allowing for secure remote access without exposing management interfaces directly to the internet
- Deployed Prometheus and Grafana with Docker Compose to visualize network bandwidth, server temperatures, and hardware health, enabling centralized diagnostics and troubleshooting
- Resolved overlapping-subnet routing conflicts by migrating to a non-conflicting LAN subnet, restoring access to homelab devices for all remote VPN clients

Distributed Command Execution & Membership Service *August 2026 - Present*

- Built the networking and cluster management layer for a distributed system of 10 Linux VMs in Go, supporting dynamic node joins, leaves, crashes, and rejoins
- Implemented peer-to-peer gossip-based failure detection with heartbeats and SWIM-style suspicion, which detects failures within 3 seconds and propagates membership changes across the system within 6 seconds
- Built a distributed command execution service using concurrent TCP connections and goroutines, enabling system-wide log querying and streamlined debugging from a single node
- Simulated an unreliable network by randomly dropping gossip messages and maintained 0 false-positive failure detections up to 20% message loss

TECHNICAL SKILLS & CERTIFICATIONS

Programming Languages: C/C++, Go, Python, SQL, Java
Platforms/Tools: Linux, OPNsense/FreeBSD, Docker, Prometheus, Grafana, WireGuard, Git
CompTIA A+ ce Certification
Issued by CompTIA *August 2025 - August 2028*

- Proficiency in hardware/software troubleshooting, networking fundamentals, and IT security
- Practical knowledge of IP addressing, network ports/protocols, and troubleshooting networks